

Review for STAT301 Final Exam

Final exam information:

- The STAT301 final exam will be on **Wednesday, August 19, 2026, 3:30pm–5:40pm (130 minutes) in Life Building 2302, in-person.**
- The format will be similar to the midterm.
- Hard copies of the exam will be distributed. You will have 130 minutes for writing your solutions, at most 20 minutes (5:40pm – 6:00pm) for uploading your solutions in a **single pdf file** to Canvas “**Assignment**” section (click “**Final Exam Solutions upload here**” link).
- You can either write your solutions on the exam paper, scan it, then upload to Canvas, or write on your laptop and then upload to Canvas.

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- The final exam is **cumulative**, covering all the materials covered throughout the term, but with more weight on the materials covered after the midterm.
- The midterm exam is **closed books/notes**. You may bring in **two letter-size sheets of notes (both sides, written or typed)**.
- **You are allowed to access Canvas/Internet only at the end of the exam for uploading your solutions.**
- You need a **simple calculator**.
- Please see more details on Canvas.

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This review only contains the materials covered after the midterm. Please review the materials covered before midterm again by yourselves.

R code will not be directly tested, but you need to understand the outputs of R code.

We will check picture ID.

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Materials covered after the midterm

- Logistic regression models
- Poisson regression models
- Model diagnostics and evaluation
- Variable selection methods (e.g., stepwise, LASSO)
- Post Inference, Prediction uncertainty

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A logistic regression model is used to model **binary response**. It can be written in three equivalent ways:

$$\log \left(\frac{P(Y = 1)}{1 - P(Y = 1)} \right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p,$$

or

$$\frac{P(Y = 1)}{1 - P(Y = 1)} = e^{\beta_0} e^{\beta_1 x_1} \cdots e^{\beta_p x_p},$$

or

$$P(Y = 1) = \frac{\exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)}{1 + \exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)},$$

where $\frac{P(Y=1)}{1-P(Y=1)}$ is called **odds of “Y=1”**.

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Assumptions for logistic regression models

- The response Y only takes two possible values, denoted by $Y = 0$ or $Y = 1$.
- The response Y follows a binomial (or Bernoulli) distribution.
- The response data $y_1, y_2, \dots, y_n$ are independent.

Note that, for a Bernoulli distribution, we have

$$E(Y) = P(Y = 1), \quad \text{Var}(Y) = E(Y)(1 - E(Y)),$$

which is very different from a normal distribution where the mean and variance are un-related.

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Parameter estimates for logistic regression models are often based on the maximum likelihood method via an iterative algorithm (so it sometimes may not converge).

Parameter β_j can be interpreted as

- log-odds of “ $Y = 1$ ” in an **additive model**, or
- odds of “ $Y = 1$ ” for e^{β_j} in a **multiplicative model**, or odds ratio for a binary predictor x_j .

Other interpretations are similar to that of a multiple linear regression model, including interpretations of **interactions** and **categorical predictors** (see review slides for the midterm).

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Statistical inference for parameter β_j can be based on a normal distribution (due to central limit theorem)

Residuals are defined in a similar way as that for a linear regression model, but we should use **Pearson residuals** or **deviance residuals**, due to the relationship between mean and variance of the response. Residual plots for logistic regression models are less useful than that for linear regression models.

A more important problem in model diagnostics is the so-called **over-dispersion problem**.

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Over-dispersion (or under-dispersion) means that the variance of the response variable in the data is larger than the variance based on the assumed Binomial/Bernoulli distribution, i.e.,

$$\text{Var}(Y) > E(Y)(1 - E(Y)).$$

Thus, the assumed Binomial/Bernoulli distribution does not hold, and the R outputs of the parameter estimates and standard errors are not correct (because they are computed based on the assumed Binomial/Bernoulli distribution).

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When there is an over dispersion problem, we can introduce a **dispersion parameter** η and assume

$$\text{Var}(Y) = \eta E(Y)(1 - E(Y)).$$

In this case, the distribution of Y is no longer Binomial or Bernoulli.

If the assumed Binomial/Bernoulli distribution holds, we should have $\eta = 1$.

We can use “quasibinomial” option in R to estimate the value of η . If η is far from 1, we should use the results from the “quasibinomial” fit.

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A Poisson regression model is used for **count responses**. It can be written as an **additive model**

$$\log(E(Y)) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p,$$

or, as a **multiplicative model**

$$E(Y) = e^{\beta_0} e^{\beta_1 x_1} \cdots e^{\beta_p x_p}.$$

Assumptions:

- The response Y is count,
- The response Y follows a Poisson distribution
- The response data $y_1, y_2, \cdots, y_n$ are independent.

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The interpretations of the regression parameters β_j , categorical predictors, and interactions, are similar to that for a logistic regression model, except that β_j measures the association between predictors and **log mean count** (additive model) or **mean count** (multiplicative model).

Statistical inference is also similar to that for a logistic regression model based on normal distributions (due to CLT).

Residual diagnostics are also similar to that of a logistic regression model, with **Pearson residuals** or **Deviance residuals**.

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If the Poisson distributional assumption holds for a Poisson regression model, we should have

$$\text{Var}(Y) = E(Y) = e^{\beta_0} e^{\beta_1 x_1} \dots e^{\beta_p x_p}.$$

In data analysis, if the observed variance of Y -data is larger than the above theoretical variance, i.e., if $\text{Var}(Y) = \eta E(Y)$ with $\eta > 1$, we have an **over-dispersion problem**. In this case, we can use the “quasipoisson” option in R for inference.

Over-dispersion problems are more common in Poisson models than in logistic models, and is more important in model diagnostics than residual plots.

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We have discussed three types of regression models: linear model, logistic model, and Poisson model. All of them can be written in a unified way as a **generalized linear model (GLM)**:

$$g(E(Y)) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p,$$

where $g(y)$ is called a **link function**, and

$L = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$ is called a **linear predictor**.

The choice of the link function depends on the type of the response variable. For linear model, we choose $g(y) = y$. For logistic model, we choose $g(y) = \log \frac{y}{1-y}$. For Poisson model, we choose $g(y) = \log(y)$.

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For any regression models, we need to consider

- **goodness-of-fit** of the model: how well does a model fit the data,
- **model/variable selection**: how to choose which predictors to be included in the model.

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For linear regression models (continuous response Y), we have the following **decomposition of variations** (measured by **sums of squares**):

$$TSS = ESS + RSS, \quad i.e., \quad \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

where TSS (total sums of squares) measures total variation in the data, ESS (explained sums of squares) measures the variation explained by the model, and RSS (residual sums of squares) measures variation in the noise.

We can then compare the magnitudes of these variations to see how useful a model is.

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The **coefficient of determination** R^2 is the proportion of total variation explained by the model:

$$R^2 = \frac{ESS}{TSS} \quad \left(= 1 - \frac{RSS}{TSS} \right).$$

It is often used to compare different models and choose predictors (models with large values of R^2 are preferred).

Another common model selection criteria is the AIC value. Models with smaller AIC values are preferred.

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The **coefficient of determination** R^2 is a good measure of how useful a model is, but its value increases with the number of predictors, so it favors larger models.

However, in data analysis, we prefer **parsimonious models** while maintaining goodness-of-fit (i.e., models with fewer predictors are preferred). Thus, a better measure is the **adjusted** R^2 :

$$\text{adjusted } R^2 = 1 - \frac{RSS/(n - p - 1)}{TSS/(n - 1)},$$

where p is the number of predictors in the model.

We can use adjusted R^2 to compare models.

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Other evaluation metrics for a linear regression model include **residuals standard error (RSE)** and **mean square error (MSE)**.

The **residuals standard error (RSE)** is defined as

$$RSE = \sqrt{RSS/(n - p - 1)}.$$

It is a good estimate of the variance of the random error e in a linear model $y = \beta_0 + \beta_1 x + e$ with $e \sim N(0, \sigma^2)$, i.e.,

$$\hat{\sigma}^2 = RSE.$$

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The **mean square error (MSE)** is defined as

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

It measures how close the predicted/fitted value $\hat{y}_i$ to the observed value y_i .

Thus, the smaller the MSE, the better the model fits the data.

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Thus, we have introduced three metrics to measure **goodness-of-fit** of a multiple linear regression model:

adjusted R^2 , RSE, and MSE

.

We usually choose a model (or predictors) with large value of adjusted R^2 or small values of RSE/MSE. However, the chosen model may or may not be **significantly** better than other models (which requires hypothesis testing such as an F-test or χ^2 -test).

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Hypothesis testing can be used to compare **nested models** to see if a model is significantly better than another one. For example, if a model with more predictors is significantly better than a smaller model with less predictors. If not, we should choose the smaller model.

The **F-test** can be used to compare any two **nested models**.

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Suppose that linear Model I has q predictors, and linear Model II has $p - q$ additional predictors than Model I (so Model I is nested within Model II), $p > q$. We wish to test

$$H_0 : \beta_{q+1} = \cdots = \beta_p = 0, \quad \text{vs} \quad H_1 : \text{not } H_0.$$

The test statistic is

$$F = \frac{(RSS_{ModelI} - RSS_{ModelII}) / (p - q)}{RSS_{ModelII} / (n - p - 1)}.$$

The p-value can then be computed based on the $F(p - q, n - p - 1)$ distribution.

The computation is usually done in R.

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The foregoing methods are designed for multiple linear regression models with continuous responses. For logistic regression or Poisson regression models, we usually compare nested models based on the **deviance**.

The deviance may be viewed as a generalization of the RSS for linear models. The lower the deviance, the better the fit.

We can compare nested models based on the **deviance test**, whose test statistic follows a χ^2 -distribution.

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Suppose that GLM Model I has q predictors, and GLM Model II has $p - q$ additional predictors than Model I (so Model I is nested within Model II), $p > q$. We wish to test

$$H_0 : \beta_{q+1} = \cdots = \beta_p = 0, \quad \text{vs} \quad H_1 : \text{not } H_0.$$

The test statistic is Deviance of Model I – Deviance of Model II. The p-value can then be computed based on the $\chi^2(p - q)$ distribution.

The computation is usually done in R.

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One of the most popular variable selection method is called **Akaike information criterion (AIC)**:

AIC = goodness-of-fit + penalty for number of predictors.

That is, we choose a model which fits the data well but it does not contain too many predictors.

The smaller the AIC value, the better the model.

Similarly, there is also a **Bayesian information criterion (BIC)**.

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When we select predictors for a regression model, a commonly used method is called the **stepwise method**. It can either do **forward selection**, **backward selection**, or both.

A forward stepwise method starts with a null model, and then it adds predictors sequentially, one at a time, based on how the (say) AIC values change.

A backward stepwise method starts with a full model, and then it deletes predictors sequentially, one at a time, based on how the (say) AIC values change.

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In these methods, a predictor is either included in the model or removed from the model (i.e., its regression coefficient is either 0 or not).

Can we do something in-between? That is, instead of put a regression coefficient for a predictor either as 0 or not, can we **shrink it** (or make the variable selection process **smaoother**)?

Yes, **LASSO (least absolute shrinkage and selection operator)** is such a method (called **regularized method**)!

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LASSO puts a penalty on the size of the regression coefficients, i.e., we estimate the parameters β_j 's to minimize

$$RSS + \lambda \sum_{j=1}^p |\beta_j|,$$

where λ is called the **penalty parameter**.

If $\lambda = 0$, the parameter estimates are just the least square estimates.

As λ increases, the coefficients β_j 's are shrunk (some may become 0, so the corresponding predictors are removed from the model).

In data analysis, the value of λ can be chosen based on the **cross-validation method**, which chooses λ to minimize MSE.

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LASSO has been used as a method to select strong predictive models. However, the regression estimates $\hat{\beta}_j$ from LASSO can be biased (i.e., $E(\hat{\beta}_j) \neq \beta_j$).

The solution is to use LASSO to select the predictors, and then use the least square method to do inference.

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Can we use the same dataset to select the predictors and also for inference? No. This may lead to inflated type I error, which is called the **post-inference problem**.

One solution is to split the dataset into two parts: one to select the model and the other one for inference. This is the idea behind the **cross-validation method**.

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When we use the fitted model for predictions, the predicted value $\hat{y}_i$ contains two sources of uncertainty: one from the usual sample-to-sample variation, and the other one from the error term e in the linear model.

This is called **prediction uncertainty**.

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Any questions before the final exam? You can either post them on Piazza or come to my office hour on Tuesday, August 19, 2pm-3pm via Zoom.

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Thank you very much!

See you on Wednesday, August 19!